NCERT Summary: Topic 11: Natural Vegetation, Wildlife & Biomes — Raw Notes

Source Integration

- NCERT Class 7 — Our Environment, Chapter 6: Natural Vegetation and Wild Life
- NCERT Class 11 — Fundamentals of Physical Geography, Chapter 14: Biodiversity and Conservation

Core Factual & Conceptual Architecture

1. World Natural Vegetation Typologies

- Climatic Determinants: Natural vegetation distribution depends primarily on temperature and moisture, and secondarily on slope and soil thickness. Vegetation changes with altitude and latitude.
- Three Broad Categories:
  - Forests: Thriving where temperature and rainfall are plentiful.
  - Grasslands: Found in regions of moderate rainfall.
  - Shrubs: Thorny scrub growing in dry arid regions.

2. Forest and Grassland Sub-Types

- Forest Classifications:
  - Tropical Evergreen (Rainforest): Hot, heavy rain year-round with dense canopies; key trees include rosewood, ebony, and mahogany.
  - Tropical Deciduous (Monsoon): Seasonal climate with trees shedding leaves in dry seasons; key trees include sal, teak, neem, and shisham.
  - Temperate Evergreen & Deciduous: Mid-latitude coastal and higher latitude forests featuring oak, pine, ash, and beech.
  - Mediterranean Vegetation: Adapted to hot dry summers and mild rainy winters with thick bark and wax-coated leaves (citrus fruits, olives, figs).
  - Coniferous Forest (Taiga): High-latitude softwood evergreen forests (chir, pine, cedar) used for paper and timber.
- Grassland Classifications:
  - Tropical Grasslands: Low-to-moderate rainfall with tall grasses up to 3 to 4 meters, known as Savanna in East Africa, Campos in Brazil, and Llanos in Venezuela.
  - Temperate Grasslands: Mid-latitude continental interiors featuring short, nutritious grass known as Pampas in Argentina, Prairies in North America, Veld in South Africa, Steppes in Central Asia, and Downs in Australia.

3. Biodiversity Architecture & Three Levels

- Definition: Biodiversity represents the variety of life forms (plants, animals, microorganisms), their genes, and forming ecosystems, evolved over 2.5 to 3.5 billion years. Richness is concentrated in the tropics.
- Three Levels of Biodiversity:
  - Genetic Diversity: Variation of genes within a species, essential for healthy breeding populations.
  - Species Diversity: Variety and number of species in a defined area, rich areas termed hotspots.
  - Ecosystem Diversity: Variety of habitat types and ecological processes within a region.

4. Importance & Loss of Biodiversity

- Three Roles of Biodiversity: Ecological (ecosystem functioning, climate regulation), economic (resources for food, pharmaceuticals, agro-biodiversity), and scientific/ethical (evolutionary insight and intrinsic right to exist).
- Causes of Biodiversity Loss: Human population growth, habitat destruction in tropical

regions, natural calamities, toxic pollutants, introduction of exotic non-native species, and illegal poaching.

5. IUCN Categories & Conservation Frameworks

- IUCN Threat Categories: Endangered species in danger of extinction (tracked via the Red List), vulnerable species likely to become endangered, and rare species with small scattered populations.
- Conservation Actions: World Conservation Strategy guidelines, India's Wildlife Protection Act 1972, 12 global mega-diversity centres, biodiversity hotspots, and the 1992 Rio Earth Summit Convention on Biodiversity.

✍ Notes & Updates